

Technical University of Denmark

Written examination: 14th December 2023, 9 AM — 1 PM.

Course name: Introduction to Machine Learning and Data Mining.

Course number: 02450.

Aids allowed: All aids permitted.

Exam duration: 4 hours.

Weighting: The individual questions are weighted equally.

The exam is multiple choice. All questions have four possible answers marked by the letters A, B, C, and D as well as the answer “Don’t know” marked by the letter E. Correct answer gives 3 points, wrong answer gives -1 point, and “Don’t know” (E) gives 0 points.

This exam only allows for electronic hand-in.

You hand in your answers at <https://eksamen.dtu.dk/>. To hand in your answers, write them in the file `answers.txt` (this file is available from the same place you downloaded this file). When you are done, upload the `answers.txt` file (and nothing else). Double-check that you uploaded the correct version of the file from your computer.

Do not change the format of `answers.txt`

The file is automatically parsed after hand-in. Do not change the file format of `answers.txt` to any other format such as `rtf`, `docx`, or `pdf`. Do not change the file structure. Only edit the portions of the file indicated by question marks.

	Attribute description	Abbrev.
x_1	Temperature measured in Celsius ($\mathbb{R}_{\geq 0}$)	Temperature (T)
x_2	Ambient pressure measured in millibar ($\mathbb{R}_{\geq 0}$)	Pressure (AP)
x_3	Relative humidity measured in percentage ($\mathbb{R}_{\geq 0}$)	Humidity (RH)
x_4	Exhaust vacuum measured in cm of mercury ($\mathbb{R}_{\geq 0}$)	Vacuum (EV)
y	Net hourly electrical energy output level	

Table 1: Description of the attributes in the Combined Cycle Power Plant (CCPP) dataset used in this exam. The dataset is used for determining the net hourly electrical energy output level. The dataset consists of $N = 9568$ observations and we consider the two classes $y = 1$ and $y = 2$ (corresponding to Low and High energy production, respectively).

Question 1. We consider the four attributes of the CCPP dataset¹ in Table 1. A scatter plot matrix for these attributes is shown in Figure 1.

Which one of the following matrices represents the empirical correlation matrix for these attributes?

A. $\hat{C} \approx \begin{bmatrix} 1.00 & 0.85 & -0.51 & -0.55 \\ 0.85 & 1.00 & -0.42 & -0.33 \\ -0.51 & -0.42 & 1.00 & 0.72 \\ -0.55 & -0.33 & 0.72 & 1.00 \end{bmatrix}$

B. $\hat{C} \approx \begin{bmatrix} 1.00 & 0.85 & -0.51 & 0.55 \\ 0.85 & 1.00 & -0.42 & -0.33 \\ -0.51 & -0.42 & 1.00 & 0.09 \\ 0.55 & -0.33 & 0.09 & 1.00 \end{bmatrix}$

C. $\hat{C} \approx \begin{bmatrix} 1.00 & 0.85 & -0.51 & -0.55 \\ 0.85 & 1.00 & -0.42 & -0.33 \\ -0.51 & -0.42 & 1.00 & 0.09 \\ -0.55 & -0.33 & 0.09 & 1.00 \end{bmatrix}$

D. $\hat{C} \approx \begin{bmatrix} 1.00 & 0.25 & -0.51 & -0.55 \\ 0.25 & 1.00 & -0.42 & -0.33 \\ -0.51 & -0.42 & 1.00 & 0.09 \\ -0.55 & -0.33 & 0.09 & 1.00 \end{bmatrix}$

E. Don't know.

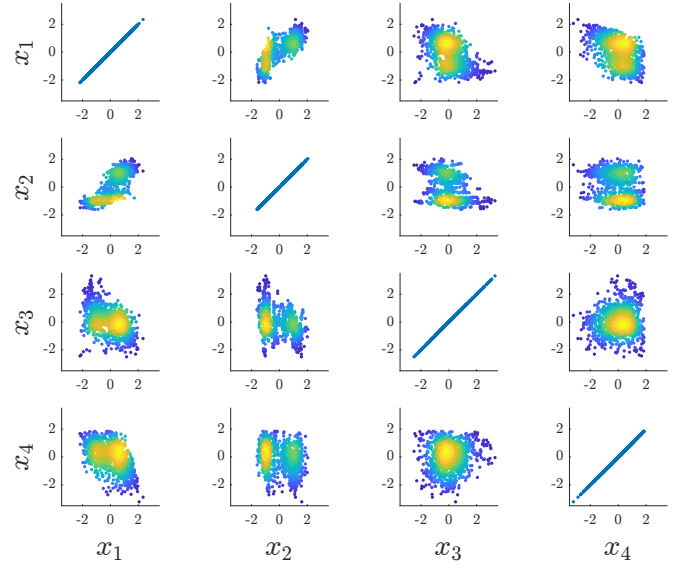


Figure 1: The scatter plot matrix for the four attributes x_1, x_2, x_3, x_4 of the CCPP dataset described in Table 1. In the off-diagonal panels, the color represents the associated density.

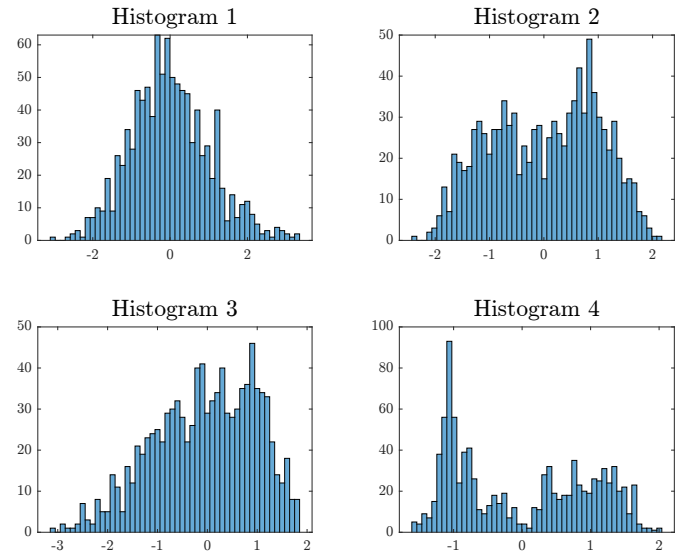


Figure 2: Histograms of the attributes in which their order has been permuted.

¹Dataset obtained from <https://archive.ics.uci.edu/dataset/294/combined+cycle+power+plant>

Question 2. We consider the histograms of the four attributes in Figure 2. Match the histograms to the attributes by using the information regarding the distribution of the points from Figure 1.

Which one of the following statements is *correct*?

- A. Histogram 1: x_3 , Histogram 2: x_4 ,
Histogram 3: x_1 , Histogram 4: x_2 .
- B. Histogram 1: x_3 , Histogram 2: x_1 ,
Histogram 3: x_4 , Histogram 4: x_2 .
- C. Histogram 1: x_2 , Histogram 2: x_3 ,
Histogram 3: x_1 , Histogram 4: x_4 .
- D. Histogram 1: x_2 , Histogram 2: x_1 ,
Histogram 3: x_3 , Histogram 4: x_2 .
- E. Don't know.

Question 3. We consider a subset of $N=5$ points from the CCPP dataset $\mathbf{X}$ where each row corresponds to an observation. We apply Principal Component Analysis (PCA) using the Singular Value Decomposition (SVD) and the resulting matrices $\mathbf{\Sigma}$, $\mathbf{V}$ are given below. The observation $\mathbf{x}_4 = [1.4 \ 1 \ 0.1 \ -2.8]^T$ is projected and then reconstructed using the first two principal components and the mean. The data matrix is

$$\mathbf{X} = \begin{bmatrix} -0.6 & -0.6 & 2.5 & -0.1 \\ -0.8 & -0.3 & -1 & 1.2 \\ -0.7 & 0.3 & -0.2 & -0.1 \\ 1.4 & 1 & 0.1 & -2.8 \\ -0.2 & 0.8 & -1.2 & 0.7 \end{bmatrix},$$

with $\tilde{X}_{ij} = X_{ij} - \frac{1}{N} \sum_{n=1}^N X_{nj}$ the centered matrix and the singular value decomposition

$$\tilde{\mathbf{X}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T, \quad \mathbf{\Sigma} \approx \begin{bmatrix} 3.7 & 0 & 0 & 0 \\ 0 & 3.04 & 0 & 0 \\ 0 & 0 & 0.56 & 0 \\ 0 & 0 & 0 & 0.48 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{V} \approx \begin{bmatrix} 0.43 & -0.26 & 0.22 & -0.84 \\ 0.17 & -0.37 & 0.81 & 0.42 \\ 0.33 & 0.88 & 0.34 & -0.01 \\ -0.82 & 0.14 & 0.42 & -0.36 \end{bmatrix}$$

Which one of the following reconstructions is *correct*?

- A. $\hat{\mathbf{x}}_4 \approx [1.35 \ 1.11 \ 0.13 \ -2.77]^T$
- B. $\hat{\mathbf{x}}_4 \approx [1.42 \ 1.15 \ 0.15 \ -2.91]^T$
- C. $\hat{\mathbf{x}}_4 \approx [1.53 \ 0.97 \ 0.19 \ -2.35]^T$
- D. $\hat{\mathbf{x}}_4 \approx [1.6 \ 0.91 \ 0.11 \ -2.69]^T$
- E. Don't know.

Question 4. We consider again the PCA result of the CCPP dataset.

Which one of the following statements is *correct*?

- A. The first two eigenvectors explain less than 90 of the total variance.
- B. With respect to the first eigenvector, low values of Temperature, Pressure, Humidity and a high value of Vacuum results in a positive projection.
- C. For every dataset, if we do not use all the principal components, then we are not able to reconstruct perfectly the given data.
- D. For every dataset it holds for the eigenvectors computed by SVD that the angle between every pair of distinct eigenvectors is 90° degrees and the Euclidean norm of every individual eigenvector is equal to one.
- E. Don't know.

Question 5. Which one of the following statements regarding logistic regression is *correct*?

- A. Consider a regularized logistic regression model for a binary classification problem. Assume that during training we over-regularize ($\lambda \rightarrow +\infty$) the weights while the intercept is not regularized. The resulting model will classify every test point to the class with the highest number of training points.
- B. We can find the optimal weights for a regularized logistic regression classifier using the formula $\mathbf{w}^* = (\tilde{\mathbf{X}}^T \tilde{\mathbf{X}} + \lambda \mathbf{I})^{-1} \tilde{\mathbf{X}}^T \mathbf{y}$.
- C. The logistic regression classifier cannot be modified to separate two classes with a nonlinear decision boundary.
- D. The logistic sigmoid function maps from a probability to a continuous value in $(-\infty, \infty)$.
- E. Don't know.

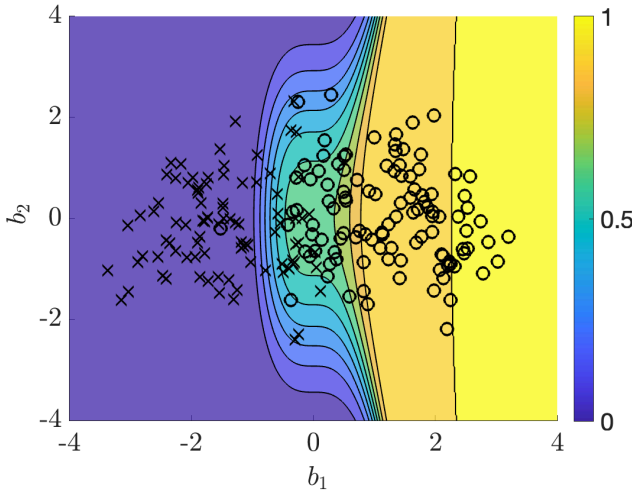


Figure 3: The logistic regression classifier, where the probability $p(y = \text{High} \mid \mathbf{b})$ is shown.

Question 6. We consider the projection of the CCPP dataset on the first two principal components, and a logistic regression classifier where we use nonlinear transformations on the projection $\mathbf{b} = [b_1 \ b_2]^T$. The resulting classifier assigns for each $\mathbf{b}$ the probability $p(y = \text{High} \mid \mathbf{b}) = \sigma(\mathbf{w}^T \tilde{\mathbf{b}})$ as shown in Figure 3.

Which one of the following combination of transformation and weight vector corresponds to this classifier?

- A. $\mathbf{w} = [0.31 \ -0.06 \ 0.07]^T$
 $\tilde{\mathbf{b}} = [1 \ b_1^2 \ b_2^3]^T$
- B. $\mathbf{w} = [0.72 \ 3.13 \ -0.25]^T$
 $\tilde{\mathbf{b}} = [1 \ b_1^3 \ b_2^2]^T$
- C. $\mathbf{w} = [0.31 \ -0.06 \ 0.07]^T$
 $\tilde{\mathbf{b}} = [1 \ b_1^3 \ b_2^2]^T$
- D. $\mathbf{w} = [0.72 \ 3.13 \ -0.25]^T$
 $\tilde{\mathbf{b}} = [1 \ b_1^3 \ b_2^2]^T$
- E. Don't know.

Question 7. Consider an artificial neural network (ANN) trained on the CCPP dataset described in Table 1 to predict the continuous attribute x_3 from the two attributes x_1 and x_2 . Suppose the neural network takes the form:

$$f(\mathbf{x}, \mathbf{w}) = h^{(2)} \left(w_0^{(2)} + \sum_{j=1}^2 w_j^{(2)} h^{(1)}([1 \ x_1 \ x_2] \mathbf{w}_j^{(1)}) \right).$$

where the activation functions are selected as $h^{(1)}(x) = \tanh(x)$ (the hyperbolic tangent activation function)

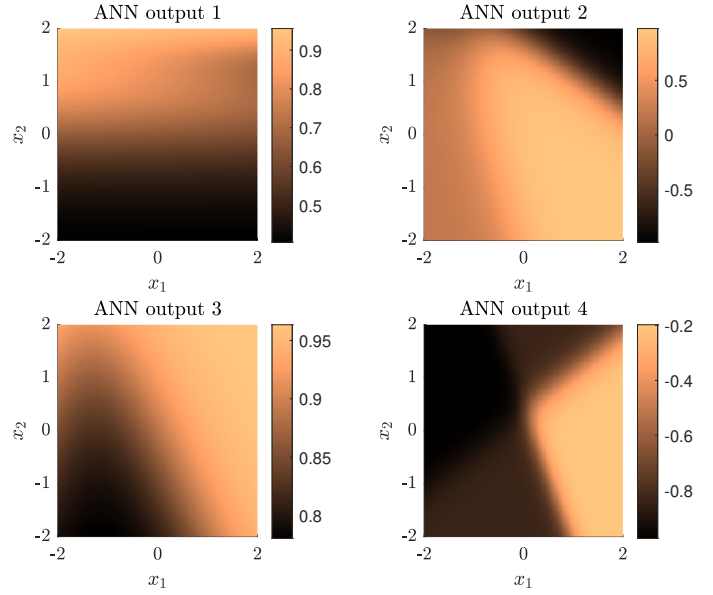


Figure 4: Suggested outputs of an ANN trained on the two attributes x_1 and x_2 from the CCPP dataset to predict the attribute x_3 .

and $h^{(2)}(x) = \tanh(x)$ (the hyperbolic tangent activation function) and the weights are given as:

$$\mathbf{w}_1^{(1)} = \begin{bmatrix} 2.2 \\ 0.7 \\ -0.3 \end{bmatrix}, \quad \mathbf{w}_2^{(1)} = \begin{bmatrix} -0.2 \\ 0.8 \\ 0.4 \end{bmatrix},$$

$$\mathbf{w}^{(2)} = \begin{bmatrix} -0.7 \\ 0.5 \end{bmatrix}, \quad w_0^{(2)} = 2.2.$$

Which one of the functions in Figure 4 will then correspond to the function f ?

- A. ANN output 1
- B. ANN output 2
- C. ANN output 3
- D. ANN output 4
- E. Don't know.

	f_1	f_2	f_3	f_4
o_1	0	1	0	0
o_2	0	1	0	0
o_3	0	1	0	1
o_4	0	1	0	0
o_5	1	1	0	1
o_6	0	1	0	1
o_7	0	0	0	1
o_8	0	0	0	1
o_9	0	0	0	1
o_{10}	0	0	0	1

Table 2: Binarized version of the CCPP dataset. Each of the features f_i are obtained by taking an attribute x_i and letting $f_i = 1$ correspond to a value x_i greater than the median (otherwise $f_i = 0$). The colors indicate classes such that the black observations $\{o_1, o_2, o_3, o_4, o_5, o_6\}$ belong to class Low and the red observations $\{o_7, o_8, o_9, o_{10}\}$ to class High.

Question 8. Consider a new, binarized observation

$$o_{11} = [0 \ ? \ ? \ 1]^T,$$

which contains two unknown values each indicated with a ?. What is the minimum and maximum achievable value of the Simple Matching Coefficient (SMC) between o_6 from Table 2 and the new observation o_{11} ?

- A. $\min \text{SMC}(o_6, o_{11}) = 0.50$
 $\max \text{SMC}(o_6, o_{11}) = 1.0$
- B. $\min \text{SMC}(o_6, o_{11}) = 0.75$
 $\max \text{SMC}(o_6, o_{11}) = 0.9$
- C. $\min \text{SMC}(o_6, o_{11}) = 0.25$
 $\max \text{SMC}(o_6, o_{11}) = 1.0$
- D. $\min \text{SMC}(o_6, o_{11}) = 0.25$
 $\max \text{SMC}(o_6, o_{11}) = 0.9$
- E. Don't know.

Question 9. We consider a subset of $N = 1000$ points from the CCPP dataset and a classifier with three different complexity levels (Model 1, Model 2, Model 3). We apply two-level cross-validation to estimate the generalization error, with 2 outer folds and 5 inner folds. In Table 3 the information for each outer fold, model, and inner fold is given, together with the

Outer fold $i=1$						
Inner folds	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$	Test
Model 1	0.12	0.21	0.22	0.23	0.15	0.24
Model 2	0.30	0.11	0.15	0.30	0.28	0.17
Model 3	0.21	0.14	0.26	0.17	0.26	0.22

Outer fold $i=2$						
Inner folds	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$	Test
Model 1	0.28	0.18	0.19	0.27	0.12	0.19
Model 2	0.16	0.20	0.27	0.30	0.25	0.16
Model 3	0.13	0.16	0.21	0.17	0.13	0.25

Table 3: The information for the two-level cross-validation. For each outer fold we are given the validation error for each model on the inner folds, together with the test error on the associated partition of the outer fold.

performance of each model on the test partition of the outer fold.

Which one the following options is the correct estimate of the generalization error?

- A. $\hat{E}^{\text{gen}} = 0.165$
- B. $\hat{E}^{\text{gen}} = 0.245$
- C. $\hat{E}^{\text{gen}} = 0.215$
- D. $\hat{E}^{\text{gen}} = 0.205$
- E. Don't know.

Question 10. Which one of the following statements regarding artificial neural networks is *correct*?

- A. An artificial neural network with linear activation functions ($h^{(i)}(x) = x$) can be written in terms of a linear regression model.
- B. We are guaranteed to find the optimal artificial neural network every time we train the model from a random initialization.
- C. The optimal number of hidden units of an artificial neural network should be selected in terms of the model that minimizes the training error.
- D. Regularizing an artificial neural network regression model by penalizing the weights using the squared Euclidean norm and a positive value λ decreases bias and increases variance.
- E. Don't know.

$p(x_1, x_3 y)$	$y = \text{Low}$	$y = \text{High}$
$x_1 = 0, x_3 = 0$	0.04	0.25
$x_1 = 0, x_3 = 1$	0.03	0.68
$x_1 = 1, x_3 = 0$	0.66	0.02
$x_1 = 1, x_3 = 1$	0.27	0.05

Table 4: The class conditional probabilities for the binarized attributes x_1, x_3 of the CCP dataset.

Question 11. Consider the Table 4 that shows the class conditional joint probability for the attributes x_1, x_3 of the CCP dataset after binarization, while the prior probabilities for the two classes are $p(y = \text{Low}) = 0.53$ and $p(y = \text{High}) = 0.47$.

What is the probability the energy production to be High when $x_1 = 0$?

- A. $p(y = \text{High} | x_1 = 0) \approx 0.92$
- B. $p(y = \text{High} | x_1 = 0) \approx 0.82$
- C. $p(y = \text{High} | x_1 = 0) \approx 0.56$
- D. $p(y = \text{High} | x_1 = 0) \approx 0.78$
- E. Don't know.

	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$
y_i	-2.9	-0.4	0.7	2.5	4.5
x_i	-3.4	-1.3	0.5	2.4	4.2

Table 5: The synthetic dataset for the regularized linear regression model.

Question 12. We consider the synthetic dataset given in Table 5 where x represents the one dimensional input and y the corresponding one dimensional output. We standardize the input before learning the optimal bias w_0^* and parameter w_1^* for a regularized linear regression model with $\lambda = 0.7$, where we consider the formulation from Section 14.1.

What is the prediction of the trained model for the input x_2 ?

- A. $\hat{y} \approx -0.76$
- B. $\hat{y} \approx -0.54$
- C. $\hat{y} \approx -0.65$
- D. $\hat{y} \approx -2.22$
- E. Don't know.

Question 13. We consider the class conditional distributions for the x_1, x_3 attributes of the CCP dataset which we model with the following multivariate normal distributions

$$p(x_1, x_3 | y = 1) = \mathcal{N}(\boldsymbol{\mu}^{(1)}, \boldsymbol{\Sigma}^{(1)}), \quad \text{where}$$

$$\boldsymbol{\mu}^{(1)} = \begin{bmatrix} 0.77 \\ -0.41 \end{bmatrix}, \quad \boldsymbol{\Sigma}^{(1)} = \begin{bmatrix} 0.29 & -0.12 \\ -0.12 & 0.55 \end{bmatrix},$$

$$p(x_1, x_3 | y = 2) = \mathcal{N}(\boldsymbol{\mu}^{(2)}, \boldsymbol{\Sigma}^{(2)}), \quad \text{where}$$

$$\boldsymbol{\mu}^{(2)} = \begin{bmatrix} -0.91 \\ 0.5 \end{bmatrix}, \quad \boldsymbol{\Sigma}^{(2)} = \begin{bmatrix} 0.32 & -0.11 \\ -0.11 & 1.12 \end{bmatrix}.$$

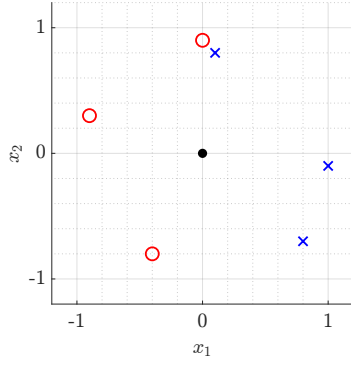
The prior probabilities for the classes Low and High are $p(y = 1) = 0.53$ and $p(y = 2) = 0.47$ respectively. We are given a new test point $\mathbf{x}_{\text{test}} = [0 \ 0.7]^T$ to classify and we consider a *Naive Bayes* approach.

Which one of the following statements is *correct*?

- A. We do not have enough information to classify the point to one of the two classes.
- B. We will classify the point to the Low class ($y = 1$).
- C. We will classify the point to the High class ($y = 2$).
- D. The point belongs in both classes with the same probability $p(y = 1 | \mathbf{x}_{\text{test}}) = p(y = 2 | \mathbf{x}_{\text{test}}) = 0.5$.
- E. Don't know.

Question 14. Which one of the following statements regarding ensemble methods is *correct*?

- A. If the accuracy of a classifier in AdaBoost is lower than 0.5, then this classifier will have no influence on the result.
- B. In AdaBoost with a regularized logistic regression classifier as the weak learner, increasing the number of iterations increases the chance of overfitting.
- C. In bagging and boosting observations are sampled without replacement, but in boosting misclassified observations are given a higher probability of being sampled in the next round.
- D. In AdaBoost the more errors a classifier makes in a given round of boosting the more the weights will be increased for the wrongly classified observations.
- E. Don't know.



	x_1	x_2
$\mathbf{o}_1$	-0.4	-0.8
$\mathbf{o}_2$	-0.9	0.3
$\mathbf{o}_3$	0	0.9
$\mathbf{o}_4$	1	-0.1
$\mathbf{o}_5$	0.8	-0.7
$\mathbf{o}_6$	0.1	0.8

Figure 5: The synthetic dataset, where the colors indicate the classes such that the red observations $\{\mathbf{o}_1, \mathbf{o}_2, \mathbf{o}_3\}$ belong to class C_1 , and the blue observations $\{\mathbf{o}_4, \mathbf{o}_5, \mathbf{o}_6\}$ belong to class C_2 .

Question 15. We are given the synthetic dataset in Figure 5 where the color indicates the class. We consider the p -distances d_2 , d_1 , d_∞ and the KNN classifier with $K = 3$ to predict the label of the test point $\mathbf{o}_{\text{test}} = [0 \ 0]^T$.

Which one of the following statements is *correct*?

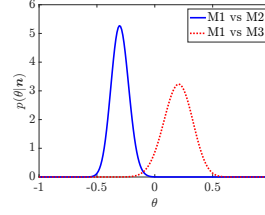
- A. d_2 and d_∞ predict the same label.
- B. d_2 and d_1 predict the same label.
- C. d_1 and d_∞ predict the same label.
- D. All the norms predict the same result
- E. Don't know.

Question 16. We consider three different classifiers (Model 1, Model 2, Model 3) and the McNemar test is used to compare M1 vs. M2 and M1 vs. M3. In Figure 6 we are given four options of Beta distributions together with the associated information for the comparisons i.e., such that M1 implies that Model 1 predicts correctly and $\overline{\text{M1}}$ that it predicts incorrectly.

For which one of the four options does the Beta distributions match the information in the table?

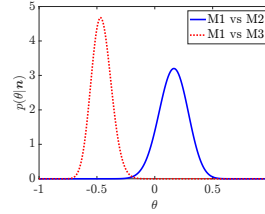
- A. Option A
- B. Option B
- C. Option C
- D. Option D
- E. Don't know.

Option A



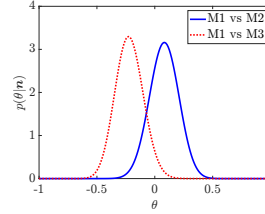
	M2	$\overline{\text{M2}}$	M3	$\overline{\text{M3}}$
M1	5	25	28	2
$\overline{\text{M1}}$	15	5	17	3

Option B



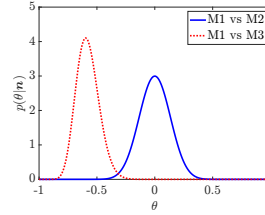
	M2	$\overline{\text{M2}}$	M3	$\overline{\text{M3}}$
M1	3	24	1	26
$\overline{\text{M1}}$	16	7	3	20

Option C



	M2	$\overline{\text{M2}}$	M3	$\overline{\text{M3}}$
M1	2	22	10	14
$\overline{\text{M1}}$	18	8	25	1

Option D



	M2	$\overline{\text{M2}}$	M3	$\overline{\text{M3}}$
M1	2	11	7	6
$\overline{\text{M1}}$	33	4	35	2

Figure 6: Four options of McNemar tests comparing M1 vs. M2 and M1 vs. M3. For each option we are given two Beta distributions together with a table that provides the information for the comparisons. Note that M1 implies that Model 1 predicts correctly and $\overline{\text{M1}}$ that it predicts incorrectly.

	o_1	o_2	o_3	o_4	o_5	o_6	o_7	o_8	o_9	o_{10}
o_1	0.0	2.6	2.8	2.1	0.8	1.7	3.9	3.8	4.1	4.2
o_2	2.6	0.0	1.8	2.3	2.7	1.4	1.9	1.7	2.0	2.2
o_3	2.8	1.8	0.0	1.2	2.9	2.1	2.2	1.6	2.3	2.0
o_4	2.1	2.3	1.2	0.0	2.4	1.8	2.6	2.5	3.0	2.9
o_5	0.8	2.7	2.9	2.4	0.0	2.0	4.2	4.0	4.4	4.5
o_6	1.7	1.4	2.1	1.8	2.0	0.0	2.4	2.6	2.7	3.0
o_7	3.9	1.9	2.2	2.6	4.2	2.4	0.0	1.2	0.6	1.3
o_8	3.8	1.7	1.6	2.5	4.0	2.6	1.2	0.0	1.0	0.6
o_9	4.1	2.0	2.3	3.0	4.4	2.7	0.6	1.0	0.0	0.9
o_{10}	4.2	2.2	2.0	2.9	4.5	3.0	1.3	0.6	0.9	0.0

Table 6: The pairwise Euclidian distances, $d(o_i, o_j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2 = \sqrt{\sum_{k=1}^M (x_{ik} - x_{jk})^2}$ between 10 observations from the CCPP dataset (recall that $M = 4$). Each observation o_i corresponds to a row of the data matrix $\mathbf{X}$ of Table 1. The colors indicate classes such that the black observations $\{o_1, o_2, o_3, o_4, o_5, o_6\}$ belong to class C_1 (corresponding to Low output), and the red observations $\{o_7, o_8, o_9, o_{10}\}$ belong to class C_2 (corresponding to High output). To avoid single features to dominate, the dataset was standardized by subtracting the mean and dividing by the standard deviation.

Question 17. Consider the observations and the pairwise distances in Table 6. At which height will groups/clusters containing o_9 and o_{10} merge in a dendrogram when using the *complete* linkage function?

- A. 1.3
- B. 0.6
- C. 0.9
- D. 1.4
- E. Don't know.

Question 18. Consider again the CCPP dataset of Table 1. Suppose we limit ourselves to $N = 6$ observations from the original dataset and furthermore suppose we consider only the features x_1 and x_4 to predict the class $y = 1$ or $y = 2$. We use a KNN classification model ($K = 1$) to this dataset and apply AdaBoost to improve the performance. After the first round of boosting, we obtain predictions and the true class labels as tabulated in Table 7. Note that $\log(x)$ consistently refers to the natural logarithm (i.e., base e). Given this information, how will the AdaBoost

Variable	y^{true}	$t = 1$
y_1	2	2
y_2	2	2
y_3	1	2
y_4	2	1
y_5	1	2
y_6	1	2

Table 7: For each of the $N = 6$ observations (first column), the table indicate the true class labels y^{true} (second column) and the predicted outputs of the AdaBoost classifier (third column) for the first round of boosting $t = 1$.

update the weights $\mathbf{w}$?

- A. [0.112 0.112 0.194 0.194 0.194 0.194]
- B. [0.25 0.25 0.125 0.125 0.125 0.125]
- C. [0.059 0.059 0.221 0.221 0.221 0.221]
- D. [0.341 0.341 0.079 0.079 0.079 0.079]
- E. Don't know.

Question 19. Consider the distances in Table 6 based on 10 observations from the CCPP dataset. The class labels C_1, C_2 (see table caption for details) will be predicted using a k -nearest neighbour classifier based on the distances given in Table 6 (ties are broken in the usual manner by considering the nearest observation from the tied classes). Suppose we use leave-one-out cross validation (i.e., the observation that is being predicted is left out) and a 3-nearest neighbour classifier (i.e., $k = 3$). What is the accuracy computed for all $N = 10$ observations?

- A. accuracy = $\frac{1}{5}$
- B. accuracy = $\frac{3}{5}$
- C. accuracy = $\frac{4}{5}$
- D. accuracy = 1
- E. Don't know.

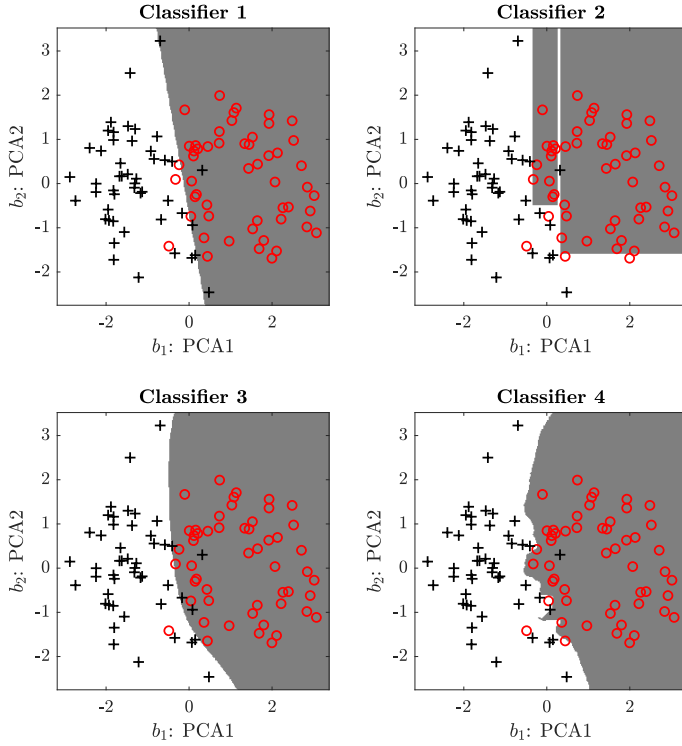


Figure 7: Decision boundaries for four different classifiers trained on the CCP dataset when projected onto the first two principal components.

Question 20. We will consider a subset of the CCP dataset (described in Table 1) after it has been projected onto the first two principal components b_1 and b_2 , thereby giving rise to a smaller two-dimensional dataset. We will consider the following four classifiers:

LREG: Logistic regression

ANN: Artificial neural network with 5 hidden units

CT: Classification tree with regular axis-aligned splits ($b_i < c$)

KNN: K-nearest neighbours with $K = 3$

Suppose the classifiers are trained on the two-dimensional dataset and the decision boundary for each of the four classifiers is given in Figure 7. Which one of the following statements is correct?

- A. Classifier 1 corresponds to **KNN**,
Classifier 2 corresponds to **ANN**,
Classifier 3 corresponds to **LREG**,
Classifier 4 corresponds to **CT**.
- B. Classifier 1 corresponds to **LREG**,

Classifier 2 corresponds to **ANN**,
Classifier 3 corresponds to **KNN**,
Classifier 4 corresponds to **CT**.

- C. Classifier 1 corresponds to **CT**,
Classifier 2 corresponds to **LREG**,
Classifier 3 corresponds to **KNN**,
Classifier 4 corresponds to **ANN**.
- D. Classifier 1 corresponds to **LREG**,
Classifier 2 corresponds to **CT**,
Classifier 3 corresponds to **ANN**,
Classifier 4 corresponds to **KNN**.
- E. Don't know.

Question 21. Consider the synthetic 1D dataset with four training observations:

$$\mathbf{X} = \{-3, -1, 5, 6\}.$$

A standard kernel density estimator (KDE) with $\sigma^2 = 0.5$ is used to determine if test observations are anomalous based on their estimated densities. An observation will be flagged as an anomaly if $p(x) < 0.015$. We test two new observations located at $x = -4$ and $x = 2$.

Determine which one of the following statements is correct.

- A. Both test observations are anomalies.
- B. The observation at $x = -4$ is *not* an anomaly but the observation at $x = 2$ is an anomaly.
- C. The obseravtion at $x = -4$ is an anomaly but the observation at $x = 2$ is *not* an anomaly.
- D. Neither of the test observations are anomalies.
- E. Don't know.

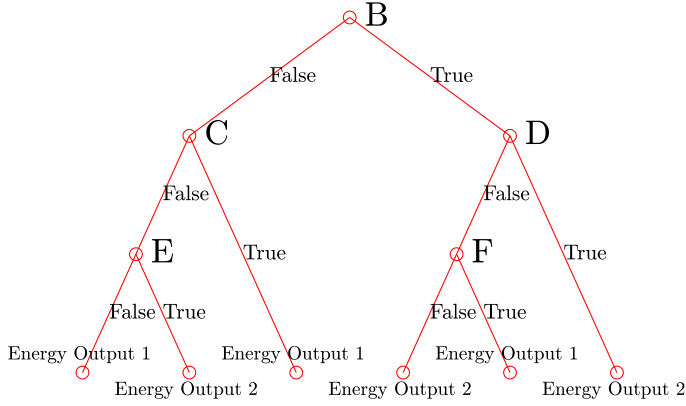


Figure 8: Structure of decision tree. The goal is to determine the splitting rules.

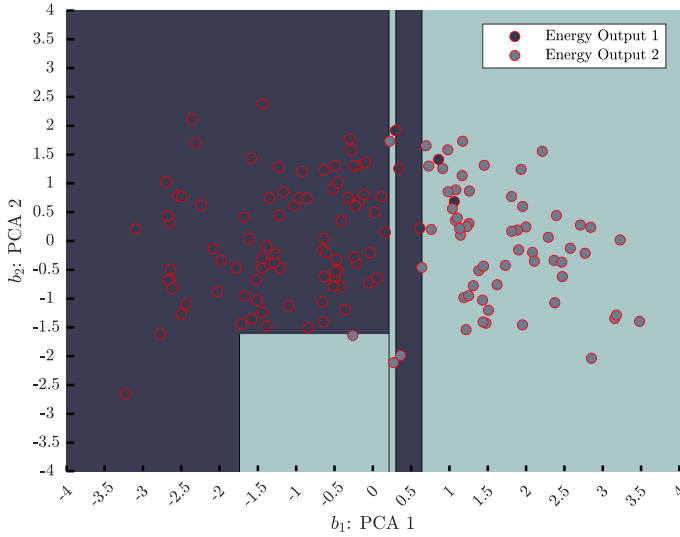


Figure 9: Classification boundary.

Question 22. We will consider the CCPP dataset projected onto the first two principal directions. Suppose we train a decision tree to predict which of the two classes an observation belongs to. Since the attributes are continuous, we will consider binary splits of the form $b_i \geq z$ for different values of i and z , where b_1, b_2 refer to the coordinates of the observations when projected onto principal directions. Suppose the trained decision tree has the form shown in Figure 8, and that according to the tree the predicted label assignment for a subset of the observations is as given in Figure 9, what is then the correct rule assignment to the nodes

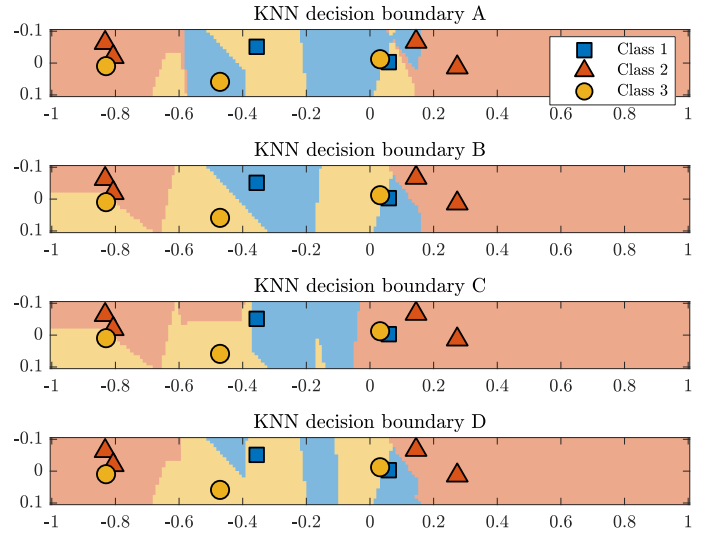


Figure 10: Decision boundaries for four KNN classifiers.

in the decision tree?

- A. **B:** $b_1 \geq 0.63$, **C:** $b_1 \geq 0.28$, **D:** $b_2 \geq -1.63$, **E:** $b_1 \geq 0.19$, **F:** $b_1 \geq -1.75$
- B. **B:** $b_1 \geq -1.75$, **C:** $b_2 \geq -1.63$, **D:** $b_1 \geq 0.28$, **E:** $b_1 \geq 0.63$, **F:** $b_1 \geq 0.19$
- C. **B:** $b_1 \geq 0.19$, **C:** $b_2 \geq -1.63$, **D:** $b_1 \geq 0.63$, **E:** $b_1 \geq -1.75$, **F:** $b_1 \geq 0.28$
- D. **B:** $b_2 \geq -1.63$, **C:** $b_1 \geq 0.63$, **D:** $b_1 \geq -1.75$, **E:** $b_1 \geq 0.28$, **F:** $b_1 \geq 0.19$
- E. Don't know.

Question 23. Consider a two-dimensional data set comprised of $N = 9$ observations shown in Figure 10. The dataset consists of three classes indicated by the blue squares (class 1), red triangles (class 2) and yellow circles (class 3). In the figure, the decision boundaries for four K -nearest neighbor classifiers (KNN) are shown. Which one of the plots correspond to the $K = 3$ nearest-neighbour classifier assuming ties are broken by assigning to the *nearest* neighbour's class?

- A. KNN decision boundary A
- B. KNN decision boundary B
- C. KNN decision boundary C
- D. KNN decision boundary D
- E. Don't know.

Question 24. A neural network classifier is trained to distinguish between two classes $y \in \{1, 2\}$ and produce class-probability $\hat{y}$ and the *receiver operator characteristic* (ROC) curve of the network when evaluated on a test set with $N = 8$ observations is shown in Figure 12. Suppose we plot the predictions on the $N = 8$ test observations by their $\hat{y}$ value along the x -axis and indicate the class labels by either a black circle (class $y = 1$) or red cross ($y = 2$), which one of the subplots in Figure 12 then corresponds to the ROC curve in Figure 11?

- A. Prediction A
- B. Prediction B
- C. Prediction C
- D. Prediction D
- E. Don't know.

Question 25. In the context of the Apriori algorithm for association rule mining, which one of the following statements accurately describes a fundamental principle or step within the algorithm?

- A. Apriori randomly samples a candidate itemset and includes it in the list of frequent itemsets if its support is $\geq \epsilon$.
- B. Apriori prunes infrequent itemsets by exploiting the downward closure property of support, where any subset of a frequent itemset must also be frequent.
- C. Apriori finds association rules with support $\geq \epsilon$ by exploiting that itemsets with $N + 1$ items *always* have lower support than itemsets with N items.
- D. Apriori uses a regression model to efficiently calculate the support of any itemset exploiting the downward closure property to find relevant inputs to the regression model.
- E. Don't know.

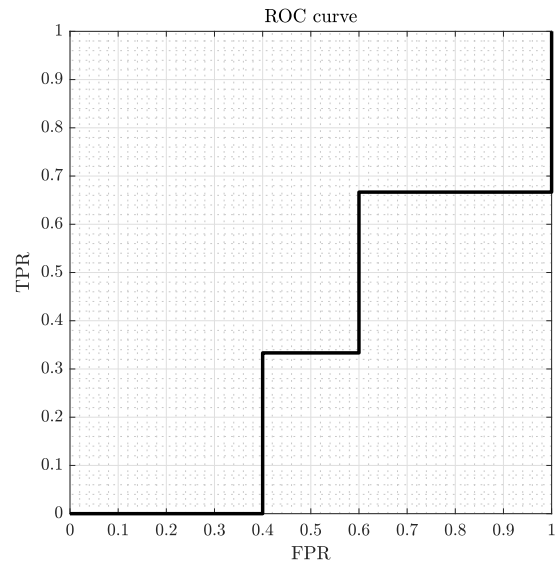


Figure 11: ROC curve for a neural network classifier, where the predictions and true class labels are one of the options in Figure 12.

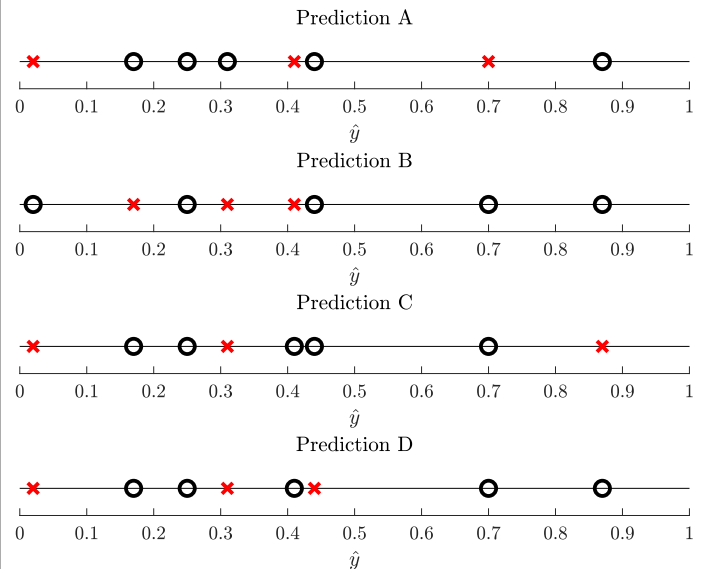


Figure 12: Four candidate predictions for the ROC curve in Figure 11. The observations are plotted horizontally, such that the position on the x -axis indicate the predicted probability $\hat{y} = p(y = 2|x)$, and the marker/color indicate the class membership, such that the black circles indicate the observation belongs to class $y_i = 1$ and red crosses to $y_i = 2$.

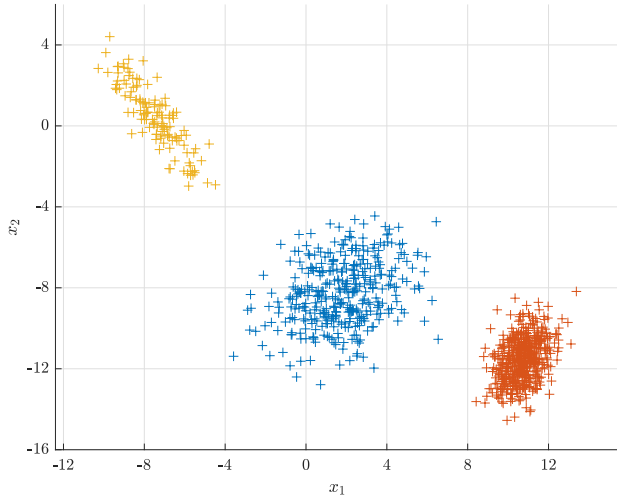


Figure 13: 1000 observations drawn from a Gaussian Mixture Model (GMM) with three clusters.

Question 26. Which one of the following statements regarding cross-validation is *correct*?

- A. The size of the test set in 10 fold cross-validation is bigger than the size of the test set in 5 fold cross-validation.
- B. Leave-one-out cross-validation gives a poor estimate of the generalization error as only one observation is part of the test set at the time.
- C. The test error obtained for the model selected based on cross-validation is a biased estimate of the generalization error.
- D. In K -fold cross-validation the trained models are more similar to each other compared to leave-one-out cross-validation because fewer data are used in each iteration for training.
- E. Don't know.

Question 27. Let $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denote the multivariate normal distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. In Figure 13 is given 1000 observations drawn from a density defined by a Gaussian Mixture Model (GMM) with three clusters. Each observation is colored in terms of which cluster it came from in the Gaussian Mixture model.

Which one of the following GMM densities was used

to generate the data?

A.

$$p(\mathbf{x}) = \frac{3}{8}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 1.7 \\ -8.4 \end{bmatrix}, \begin{bmatrix} 3.0 & 0.8 \\ 0.8 & 2.1 \end{bmatrix}\right) + \frac{1}{2}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 10.7 \\ -11.5 \end{bmatrix}, \begin{bmatrix} 0.6 & 0.3 \\ 0.3 & 1.1 \end{bmatrix}\right) + \frac{1}{8}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} -7.4 \\ 0.3 \end{bmatrix}, \begin{bmatrix} 1.3 & -1.5 \\ -1.5 & 2.3 \end{bmatrix}\right)$$

B.

$$p(\mathbf{x}) = \frac{1}{8}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 1.7 \\ -8.4 \end{bmatrix}, \begin{bmatrix} 1.3 & -1.5 \\ -1.5 & 2.3 \end{bmatrix}\right) + \frac{3}{8}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 10.7 \\ -11.5 \end{bmatrix}, \begin{bmatrix} 0.6 & 0.3 \\ 0.3 & 1.1 \end{bmatrix}\right) + \frac{1}{2}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} -7.4 \\ 0.3 \end{bmatrix}, \begin{bmatrix} 3.0 & 0.8 \\ 0.8 & 2.1 \end{bmatrix}\right)$$

C.

$$p(\mathbf{x}) = \frac{1}{8}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 1.7 \\ -8.4 \end{bmatrix}, \begin{bmatrix} 3.0 & 0.8 \\ 0.8 & 2.1 \end{bmatrix}\right) + \frac{1}{2}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 10.7 \\ -11.5 \end{bmatrix}, \begin{bmatrix} 0.6 & 0.3 \\ 0.3 & 1.1 \end{bmatrix}\right) + \frac{3}{8}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} -7.4 \\ 0.3 \end{bmatrix}, \begin{bmatrix} 1.3 & -1.5 \\ -1.5 & 2.3 \end{bmatrix}\right)$$

D.

$$p(\mathbf{x}) = \frac{3}{8}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 1.7 \\ -8.4 \end{bmatrix}, \begin{bmatrix} 1.3 & -1.5 \\ -1.5 & 2.3 \end{bmatrix}\right) + \frac{1}{2}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 10.7 \\ -11.5 \end{bmatrix}, \begin{bmatrix} 3.0 & 0.8 \\ 0.8 & 2.1 \end{bmatrix}\right) + \frac{1}{8}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} -7.4 \\ 0.3 \end{bmatrix}, \begin{bmatrix} 0.6 & 0.3 \\ 0.3 & 1.1 \end{bmatrix}\right)$$

E. Don't know.